Bonus Case

Team: _____

```
>>> print(0.1 + 0.2)
0.30000000000000004
```

Finished all four cases? These go a step further. Use what you learned about bits, rounding and colors.

1. Python prints `0.30000000000000004` for `0.1 + 0.2`. Explain why, using what you learned in Case 1.

2. A 6-bit system has the place values below. What is the closest value to **12.6** it can store? Fill in the bits.

Place value	8	4	2	1	½	¼	Value
Store 12.6							

How far off is it? _____

3. A 10 × 10 image can use 64 different colors. What is the smallest possible size in bits, ignoring the header? Show your work.

4. Design challenge: invent a 2-bit-per-pixel grayscale code (4 shades). Write your key, then encode a 3 × 3 image of your choice.

KEY IDEA A fixed number of bits always means a **limit**: on how big, how precise, or how colorful the data can be.